

Assignment No. 10

Title: Custom Application (Mini Project) using Sales force Cloud.

Problem Statement: Design and develop custom Application (Mini Project) using Sales force Cloud.

Objective: To develop custom application (Mini Project) using salesforce cloud

Outcome: Application in Salesforce.com using Apex programming Language.

Theory:

Salesforce is a cloud-based Customer Relationship Management (CRM) platform that allows users to create custom applications using declarative tools and programmatic logic. Apex is a strongly typed, object-oriented programming language used in Salesforce to execute business logic on the Salesforce platform.

Custom objects in Salesforce function similarly to database tables. Each custom object stores records and fields. In Apex, custom objects are referenced using the API name followed by __c. For example, a custom object named "Student" becomes Student__c in Apex.

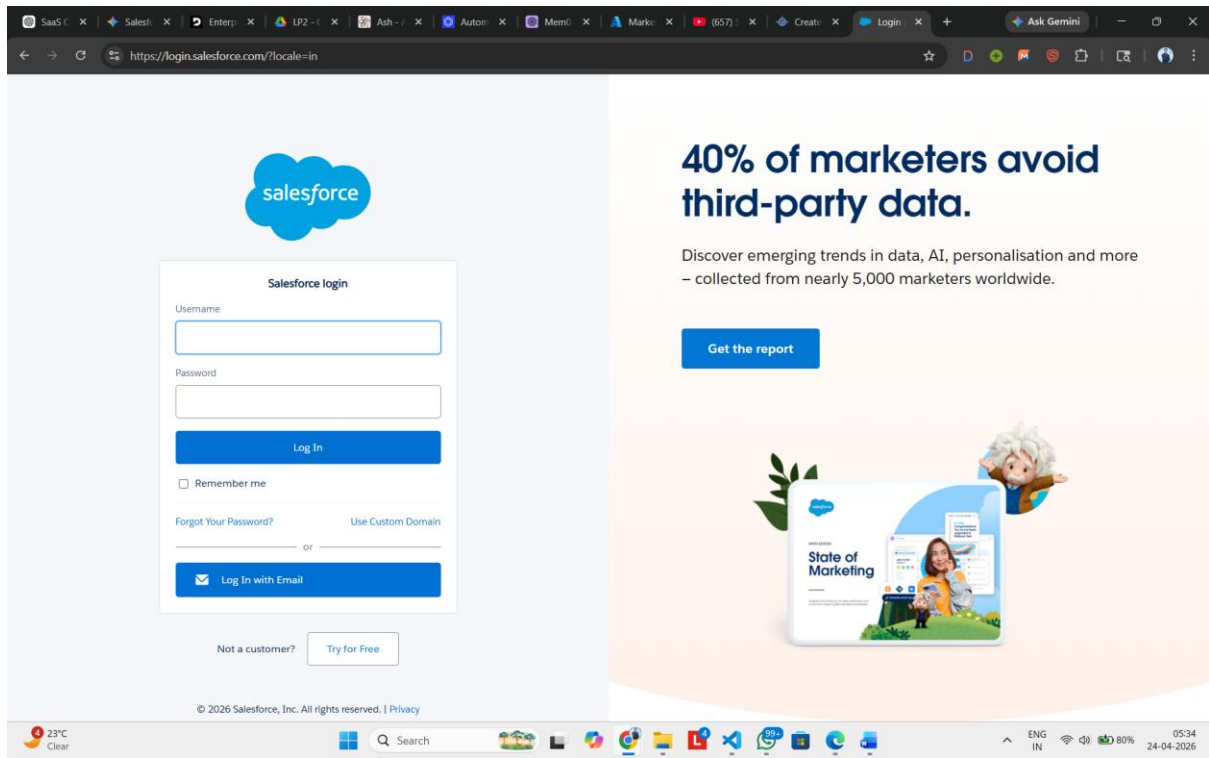
Apex supports database operations using DML (Data Manipulation Language) statements such as insert, update, and delete. Data retrieval is performed using SOQL (Salesforce Object Query Language), which is similar to SQL but specifically designed for Salesforce objects.

The Developer Console provides an environment to write, execute, and debug Apex code.

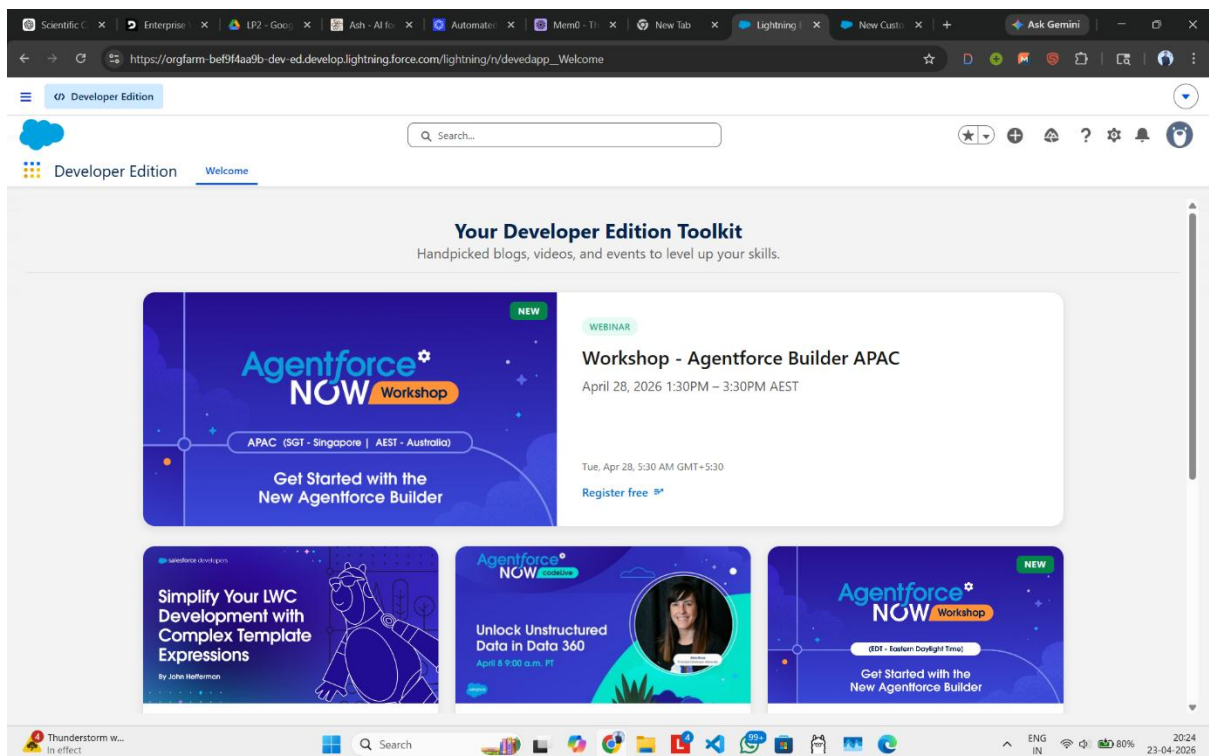
The "Execute Anonymous" feature allows developers to test Apex code without creating a permanent method

Procedure:

Step 1: Signup or login



Step 2: After this you will reach the Dashboard



Step 3: Create custom object for Task Manager

Go to: Setup → Object Manager → Create → Custom Object

The screenshot shows the 'New Custom Object' page in Salesforce Setup. The page title is 'New Custom Object'. Below the title, there is a message: 'Permissions for this object are disabled for all profiles by default. You can enable object permissions in permission sets or by editing custom profiles. Tell me more! Don't show this message again.' The main section is 'Custom Object Definition Edit', which includes a 'Custom Object Information' tab. This tab contains fields for 'Label' (with an example 'Account'), 'Plural Label' (with an example 'Accounts'), 'Starts with vowel sound' (a checkbox), 'Object Name' (with an example 'Account'), 'Description', 'Context-Sensitive Help Setting' (with options to open the standard Salesforce.com Help & Training window or a Visualforce page), and 'Content Name' (set to 'None'). Below this is the 'Enter Record Name Label and Format' section, which explains that the record name appears in page layouts, key lists, related lists, lookups, and search results. The browser's address bar shows the URL: 'https://orgfarm-bef9f4aa9b-dev-ed.develop.my.salesforce.com/lightning/setup/ObjectManager/new'. The taskbar at the bottom shows the Windows taskbar with various application icons and the system clock displaying 20:23 on 23-04-2026.

Step 4: Create the fields Description, Priority, Status and Due Date

The screenshot shows the 'Fields & Relationships' page for the 'Task' object in Salesforce Setup. The page title is 'Task'. The left sidebar shows a list of tabs: 'Details', 'Fields & Relationships' (selected), 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', 'Related Lookup Filters', 'Restriction Rules', 'Scoping Rules', 'Object Access', 'Triggers', and 'Flow Triggers'. The main content area is titled 'Fields & Relationships' and shows '4 Items, Sorted by Field Label'. There are buttons for 'New', 'Deleted Fields', 'Field Dependencies', and 'Set History Tracking'. A table lists the fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Task Name	Name	Text(80)		✓

The browser's address bar shows the URL: 'https://orgfarm-bef9f4aa9b-dev-ed.develop.my.salesforce.com/lightning/setup/ObjectManager/01dM00000Lphw/FieldsAndRelationships/view'. The taskbar at the bottom shows the Windows taskbar with various application icons and the system clock displaying 20:27 on 23-04-2026.

Scientific Calc x Enterprise Vo x LP2 - Google x Ash - AI for M x Automated p x Mem0 - The x Lightning Exp x Task | Salesforce x Ask Gemini

https://orgfarm-bef9f4aa9b-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01dM00000EphxJ/FieldsAndRelationships/new

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Task

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

- ☐ Checkbox
- ☐ Currency
- ☐ Date
- ☐ Date/Time
- ☐ Email
- ☐ Geolocation
- ☐ Number
- ☐ Percent
- ☐ Phone
- ☐ Picklist
- ☐ Picklist (Multi-Select)
- ☐ Text
- ☐ Text Area
- ☒ Text Area (Long)
- ☐ Text Area (Rich)
- ☐ Text (Encrypted)
- ☐ Time
- ☐ URL

Allows users to select a true (checked) or false (unchecked) value.

Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.

Allows users to enter a date or pick a date from a popup calendar.

Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the pop-up, that date and the current time are entered into the Date/Time field.

Allows users to enter an email address, which is validated to ensure proper format. If this field is specified for a contact or lead, users can choose the address when clicking Send an Email. Note that custom email addresses cannot be used for mass emails.

Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.

Allows users to enter any number. Leading zeros are removed.

Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.

Allows users to enter any phone number. Automatically formats it as a phone number.

Allows users to select a value from a list you define.

Allows users to select multiple values from a list you define.

Allows users to enter any combination of letters and numbers.

Allows users to enter up to 255 characters on separate lines.

Allows users to enter up to 131,072 characters on separate lines.

Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.

Allows users to enter any combination of letters and numbers and store them in encrypted form.

Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:50.600" are all valid times for this field.

Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next Cancel

33°C Mostly cloudy

Search

ENG IN 80% 20:28 23-04-2026

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https://orgfarm-bef9f4aa9b-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01dM00000EphxJ/FieldsAndRelationships/new

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER

Task

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

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Object Access

Triggers

Flow Triggers

Task

New Custom Field

Step 2. Enter the details

Step 2 of 4

Previous Next Cancel

Field Label Description

Length 32,768 (Max 131,072)

Visible Lines 3

Field Name Description

Description

Help Text

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Default Value Show Formula Editor

Use formula syntax. Enclose text and picklist value API names in double quotes ("The API"). Include numbers without quotes (25), show percentages as decimals (.18), and express date calculations in the standard format: {Today() + 7}. To

Einstein

33°C Mostly cloudy

Search

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Step 5: Add Validation Rule

The screenshot shows the Salesforce Setup interface in the Developer Edition. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, Object Access, Triggers, Flow Triggers, Validation Rules (highlighted), and Conditional Field Formatting. The main content area is titled 'Validation Rule Edit' and includes fields for Rule Name (No_Past_Date), Active status, and Description. Below these is the 'Error Condition Formula' section with an example formula: `Discount_Percent__c > 30`. The 'Error Message' section shows an example message: 'Discount percent cannot exceed 30%'. The 'Error Location' section shows the message will appear at the top of the page or below a specific field on the page.

Step 6: Create Apex Trigger

The screenshot shows the Salesforce Developer Console in Google Chrome. The top bar displays the URL: `https://orgfarm-be9f4aa9b-dev-ed.develop.my.salesforce.com/ui/common/apex/debug/ApexCSPage?action=selectExtent&text=apextrigger`. The main editor shows the code for `TaskTrigger.apxt`. The code is as follows:

```
1 trigger TaskTrigger on Task__c (before insert, before update) {
2     for(Task__c t : Trigger.new){
3         if(t.Due_Date__c != null && t.Due_Date__c < Date.today()){
4             t.Status__c = 'Pending';
5         }
6     }
7 }
```

The bottom of the console shows a table with columns: User, Application, Operation, Time, Status, Read, and Size. The table is currently empty.

Step 7: Create App 'Task Manager'

New Lightning App

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name

* Developer Name

Description

App Branding

Image

Primary Color Hex Value

☐ Org Theme Options
Use the app's image and color instead of the org's custom theme

App Launcher Preview

App Name	Developer Name	Description	Created Date	Theme	Status
Developer Edition	Developer_Edition	Welcome to your Developer Edition Org.	4/15/2026, 6:06 PM	Lightning (Managed)	✓
Digital Experiences	SalesforceCMS	Manage content and media for all of your sites.	4/15/2026, 5:41 PM	Lightning	✓
Lightning Usage App	LightningInstrumentation	View Adoption and Usage Metrics for Lightning Experience	4/15/2026, 5:41 PM	Lightning	✓

Step 8: Add Tab 'Task'

SETUP

Setup Tabs

New Custom Object Tab

Step 1. Enter the Details Step 1 of 3

Choose the custom object for this new custom tab. Fill in other details.

New Custom Object Tab

Select an existing custom object or create a new custom object now.

Object

Tab Style

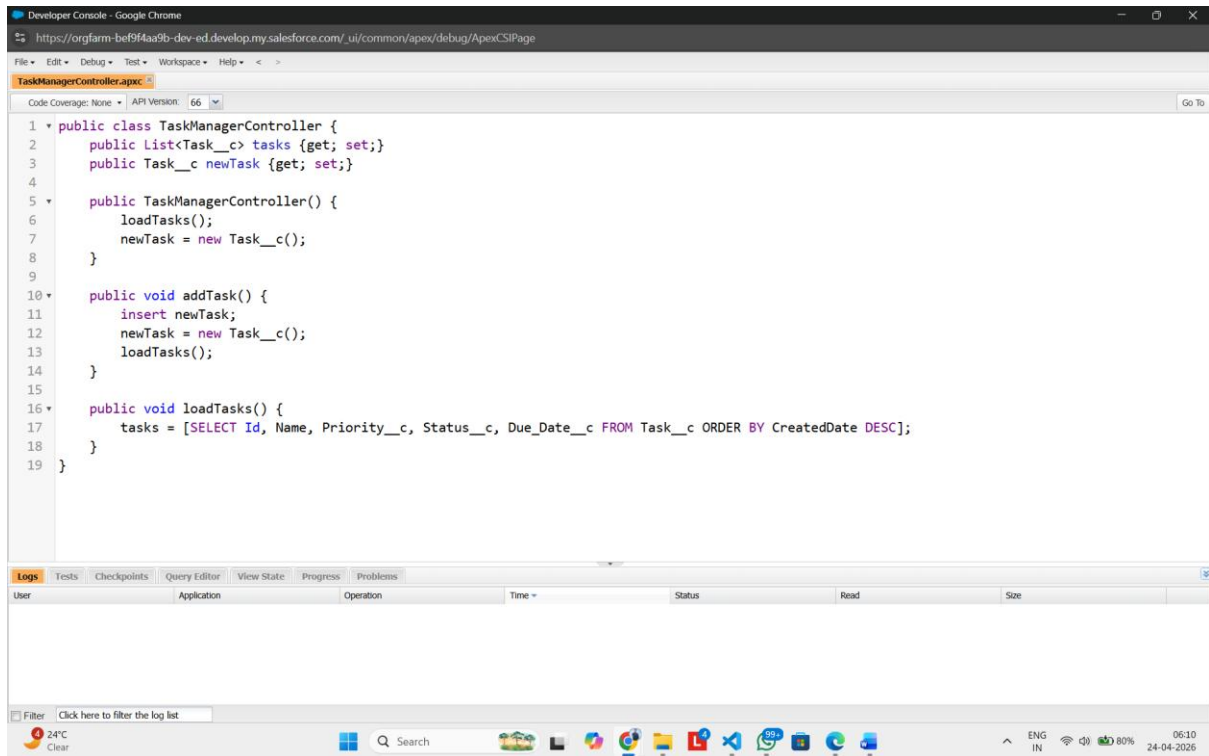
(Optional) Choose a Home Page Custom Link to show as a splash page the first time your users click on this tab.

Splash Page Custom Link

Enter a short description.

Description

Step 9: Create Apex Controller class

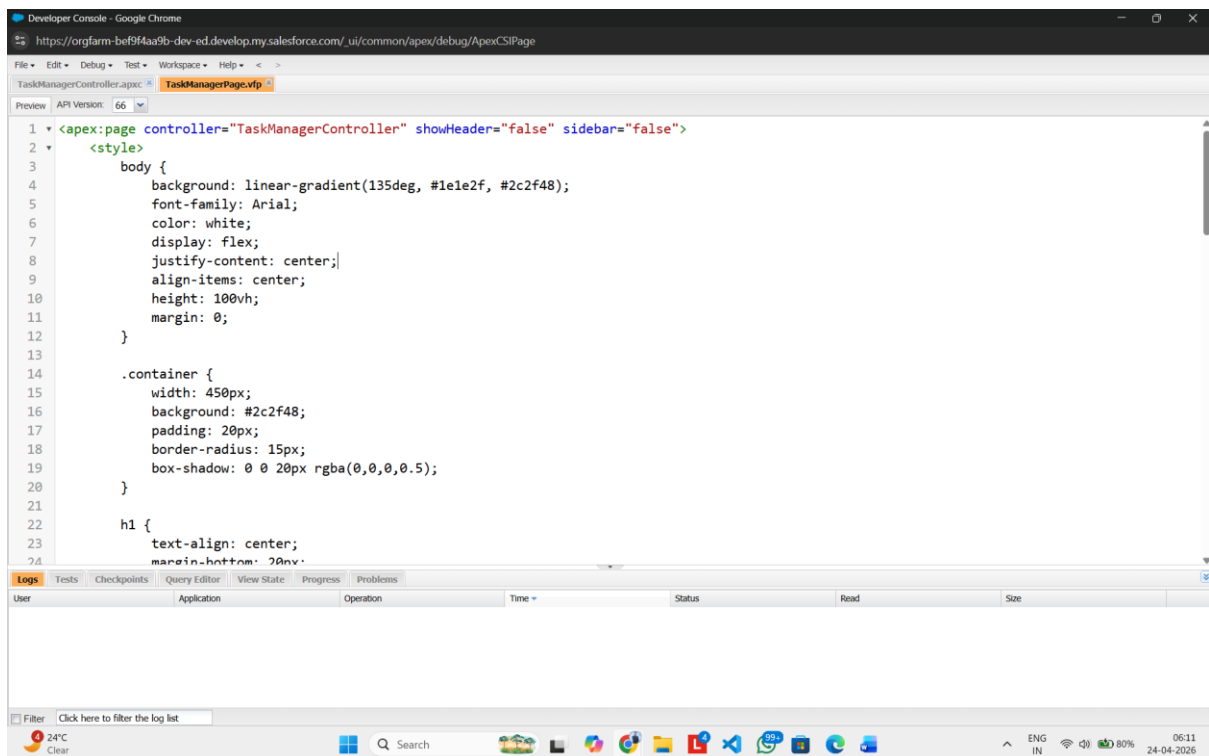


The screenshot shows the Salesforce Developer Console with the Apex class `TaskManagerController.apex` open. The class contains the following code:

```
1 public class TaskManagerController {
2     public List<Task__c> tasks {get; set;}
3     public Task__c newTask {get; set;}
4
5     public TaskManagerController() {
6         loadTasks();
7         newTask = new Task__c();
8     }
9
10    public void addTask() {
11        insert newTask;
12        newTask = new Task__c();
13        loadTasks();
14    }
15
16    public void loadTasks() {
17        tasks = [SELECT Id, Name, Priority__c, Status__c, Due_Date__c FROM Task__c ORDER BY CreatedDate DESC];
18    }
19 }
```

The console interface includes tabs for Logs, Tests, Checkpoints, Query Editor, View State, Progress, and Problems. The bottom status bar shows the system time as 06:10 on 24-04-2026.

Step 10: Create VisualForce Page

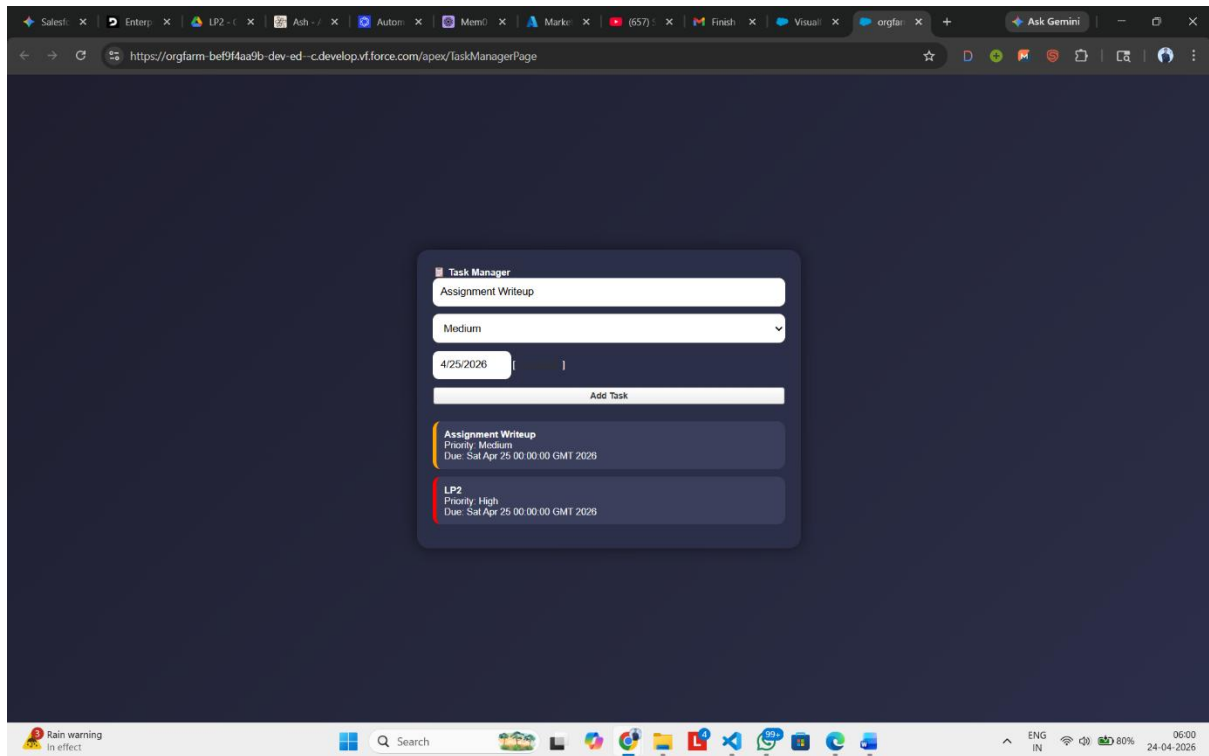


The screenshot shows the Salesforce Developer Console with the VisualForce page `TaskManagerPage.vfp` open. The page contains the following code:

```
1 <apex:page controller="TaskManagerController" showHeader="false" sidebar="false">
2     <style>
3         body {
4             background: linear-gradient(135deg, #1e1e2f, #2c2f48);
5             font-family: Arial;
6             color: white;
7             display: flex;
8             justify-content: center;
9             align-items: center;
10            height: 100vh;
11            margin: 0;
12        }
13
14        .container {
15            width: 450px;
16            background: #2c2f48;
17            padding: 20px;
18            border-radius: 15px;
19            box-shadow: 0 0 20px rgba(0,0,0,0.5);
20        }
21
22        h1 {
23            text-align: center;
24            margin-bottom: 20px;
25        }
26    </style>
27 </apex:page>
```

The console interface includes tabs for Logs, Tests, Checkpoints, Query Editor, View State, Progress, and Problems. The bottom status bar shows the system time as 06:11 on 24-04-2026.

Step 11: Preview App



Conclusion: Hence, we have successfully created an custom application on sales force cloud.